Haskell: The Purely Functional Programming Language

Haskell, an open source programming language, is the outcome of 20 years of research. It has all the advantages of functional programming and an intuitive syntax based on mathematical notation. This article flags off a series in which we will explore Haskell at length.

Haskell is a statically typed, general-purpose programming language. Code written in Haskell can be compiled and also used with an interpreter. The static typing helps detect plenty of compile time bugs. The type system in Haskell is very powerful and can automatically infer types. Functions are treated as first-class citizens and you can pass them around as arguments. It is a pure functional language and employs lazy evaluation. It also supports procedural and strict evaluation, similar to other programming paradigms.

Haskell code is known for its brevity and is very concise. The latest language standard is Haskell 2010. The language supports many extensions, and has been evoking widespread interest in the industry due to its capability to run algorithms on multi-core systems. It has support for concurrency because of the use of software transactional memory. Haskell allows you to quickly create prototypes with its platform and tools. Hoogle and Hayoo API search engines are available to query and browse the list of Haskell packages and libraries. The entire set of Haskell packages is available in Hackage.

The Haskell platform contains all the software required to get you started on it. On GNU/Linux, you can use your distribution package manager to install the same. On Fedora, for example, you can use the following command:

```
# yum install haskell-platform
```

On Ubuntu, you can use the following:

```
# apt-get install haskell-platform
```

On Windows, you can download and run *HaskellPlatform-2013.2.0.0-setup.exe* from the Haskell platform website and follow the instructions for installation.

For Mac OS X, download either the 32-bit or 64-bit *.pkg* file, and click on either to proceed with the installation.

The most popular Haskell interpreter is the Glasgow Haskell Compiler (GHC). To use its interpreter, you can run *ghci* from the command prompt on your system:

```
$ ghci
GHCi, version 7.6.3: http://www.haskell.org/ghc/ :? for help
Loading package ghc-prim ... linking ... done.
Loading package integer-gmp ... linking ... done.
Loading package base ... linking ... done.
Prelude>
```

The *Prelude* prompt indicates that the basic Haskell library modules have been imported for your use.

To exit from GHCi, type *:quit* in the *Prelude* prompt:

```
Prelude> :quit
Leaving GHCi.
```

The basic data types used in Haskell are discussed below.

A *Char* data type is for a Unicode character. You can view the type using the command *:type* at the GHCi prompt:

```
Prelude> :type 's'
's' :: Char
```